

PREDICTION OF HOURLY PM_{2.5} CONCENTRATIONS USING MACHINE LEARNING TECHNIQUES IN BUCARAMANGA, COLOMBIA

PREDICCIÓN DE CONCENTRACIONES HORARIAS DE PM_{2.5} MEDIANTE TÉCNICAS DE APRENDIZAJE AUTOMÁTICO EN BUCARAMANGA, COLOMBIA

Adrian Felipe Vargas Rojas^[1], Jorman Alexis Muñoz Anacona^[2], Saulo Quiñones Gongora^[3], Manuel Castillo Rosales^[4], Luis David Hurtado Caicedo^[5]

ABSTRACT

This project develops a predictive system for estimating hourly PM_{2.5} concentrations in Bucaramanga, Colombia, using machine learning techniques applied to historical air quality and meteorological data. The dataset includes variables such as temperature, humidity, radiation, precipitation, wind patterns, and particulate matter (PM_{2.5} and PM₁₀). Three supervised regression algorithms were implemented and evaluated—Random Forest Regressor, AdaBoost Regressor, and Ridge Regression (L2)—using an identical train-test split to ensure fair comparison. Model performance was assessed through RMSE, MAE, and R² metrics, complemented by computational complexity analysis. Random Forest achieved the best predictive accuracy (R² = 0.960 on the test set), demonstrating strong robustness to noise and effective modeling of nonlinear relationships. AdaBoost presented intermediate performance with higher sensitivity to outliers, while Ridge Regression offered minimal computational cost but limited predictive capability. The final selected model (Random Forest) was integrated into a desktop-friendly executable GUI to support real-time environmental monitoring and early-warning applications.

Keywords: [PM_{2.5}], [prediction], [machine learning], [regression models], [air quality], [environmental monitoring].

Introduction

Atmospheric pollution is one of the leading environmental threats to global public health, causing 7 million premature deaths each year [5]. In particular, fine particulate matter (PM_{2.5}) stands out as one of the most harmful pollutants, as its particles can penetrate deep into the respiratory system and trigger cardiovascular and respiratory conditions [1]. In intermediate cities such as Bucaramanga, located in the department of Santander, Colombia, this issue becomes especially relevant due to factors such as rapid urban growth, high vehicle density, and topographic conditions that favor the accumulation of pollutants.

Despite its importance, current air-quality monitoring systems are reactive in nature, as they only allow authorities to alert the population once pollutant concentrations have already reached potentially hazardous levels. These limitations reduce the ability of environmental agencies and citizens to plan preventive actions, constraining decision-making based on anticipation.

In this context, the present project seeks to shift this paradigm toward a predictive approach by leveraging the potential of machine-learning techniques. Specifically, it proposes the

development of a model capable of forecasting PM_{2.5} concentrations.

The aim of this research is not only to improve the ability to anticipate PM_{2.5} behavior in Bucaramanga, but also to provide a tool that facilitates the formulation of more efficient and timely environmental-management strategies grounded in scientific evidence.

OBJECTIVES

Main Objective

Develop a predictive model based on machine-learning techniques to estimate the future hourly concentration (one hour ahead, $t+1$) of fine particulate matter (PM_{2.5}) at a monitoring station in the city of Bucaramanga (Santander). The model will rely on meteorological and air-quality variables, with the purpose of strengthening air-pollution management and control in the city.

Secondary Objectives

- Collect, clean, and structure the historical data from the monitoring station, incorporating meteorological



Attribution 4.0 International (CC BY 4.0) Share Adapt

variables (temperature, humidity, radiation, rainfall, wind direction and speed) and associated pollutants ($PM_{2.5}$ and PM_{10}) with hourly resolution.

- Develop and train supervised machine-learning models for short-term (one hour ahead) $PM_{2.5}$ prediction, evaluating their performance using statistical error and fit metrics.
- Implement a graphical user interface (GUI) that integrates the predictive model into an executable (.exe) file, enabling easy consultation and application in monitoring or environmental-management scenarios.

PROBLEM STATEMENT

Phenomenon: Dynamics of Fine Particulate Matter ($PM_{2.5}$) Pollution

The phenomenon under study is the hourly concentration of fine particulate matter ($PM_{2.5}$) in the urban area of Bucaramanga, Colombia. $PM_{2.5}$ consists of aerosols with a diameter less than or equal to 2.5 micrometers, which allows them to penetrate deep into the human respiratory system and enter the bloodstream, making them one of the most critical atmospheric pollutants for public health. The presence and variation of these particles in the atmosphere are not random; rather, they are governed by a complex system of factors that include:

- Emission sources: Vehicular traffic, industries, biomass burning, among others.
- Meteorological conditions: Temperature, humidity, wind speed and direction, and solar radiation influence the dispersion, chemical transformation, and accumulation of pollutants.
- Temporal patterns: Concentrations exhibit predictable cyclical variations associated with the time of day (e.g., rush hours), day of the week (e.g., differences between weekdays and weekends), and season of the year (e.g., rainfall patterns).

Problem to Address: The Need for Accurate and Timely Prediction

The core issue lies in the reactive nature of current air-quality surveillance systems. These systems measure and report past and present concentrations, which means they can only generate alerts once air quality has already deteriorated. This limitation prevents the implementation of preventive and proactive measures to protect the population, especially vulnerable groups.

To shift this paradigm, the development of short-term forecasting tools is essential. While deterministic models (air-quality models based on atmospheric chemistry and physics) do exist, they are often computationally expensive and require detailed emissions data that are not always available.

Proposed Framework: A Machine-Learning Approach

This work addresses this gap by proposing a predictive model based on machine-learning techniques. The goal is to leverage historical monitoring data to learn the complex, non-linear patterns linking recent conditions (pollutants and meteorology) to future $PM_{2.5}$ concentrations.

The specific problem to be tackled is the prediction of hourly $PM_{2.5}$ concentration one hour ahead ($t+1$). This forecasting horizon is short enough to enable high accuracy and long enough to allow informed decision-making, positioning it as a key component for the development of low-cost, high-efficiency early-warning systems in urban environments. Preliminary results, with a coefficient of determination (R^2) of 0.983, validate the feasibility and strong predictive potential of our methodology.

PREVIOUS WORK

RAQ: Random Forest Approach for Predicting Air Quality in Urban Sensing Systems

This study proposed the RAQ (Random Forest Air Quality) model, based on the Random Forest algorithm, to predict the Air Quality Index (AQI) in urban environments. Real-time environmental sensor data were used to estimate pollutant concentrations such as $PM_{2.5}$ and NO_2 .

The results showed that the ensemble approach significantly reduced prediction variance and outperformed individual methods such as Linear Regression and SVM. However, the model exhibited limitations in generalizing to areas with different meteorological conditions, due to the lack of extended geospatial and temporal variables [2].

Deep Distributed Fusion Network for Air Quality Prediction

The study introduces a Deep Distributed Fusion Network (DDFN) to predict air quality in large metropolitan areas. It combines meteorological, traffic, and industrial-emissions data through multiple neural layers that integrate spatial and temporal information.

The model improved prediction accuracy by 15% compared to traditional models such as ARIMA and SVM. However, its high complexity and computational demands make its deployment challenging in real-time systems or environments with limited resources [3].

Air Quality Index Prediction Using Simple Machine Learning Algorithms

This study compared classical machine-learning methods (k-Nearest Neighbors, Decision Trees, SVM, and ANN) for predicting AQI levels in urban environments. The authors showed that an artificial neural network (ANN) achieved an accuracy of 92.3%, outperforming the other algorithms [4].

The work highlights that even simple models can deliver competitive performance, but it warns about their limited ability to capture complex non-linear relationships or seasonal environmental changes, which restricts their applicability for long-term predictions.

A Machine Learning Tool to Forecast PM₁₀ Level

One of the earliest approaches to applying SVM and ANN to the prediction of atmospheric pollutants (PM₁₀). The model was trained with meteorological data and particle concentrations to forecast future PM₁₀ levels.

The results showed that the ANN achieved a specificity of 79%, while SVM reached 80% with a minimal false-positive rate (0.13%). However, the study relied on a temporally limited dataset, preventing it from capturing seasonal variations and reducing its short-term predictive capability [6].

DATA ANALYSIS AND PREPROCESSING

In the first stage, hourly air-quality records were extracted using the Socrata API from the Datos Abiertos de Colombia platform.

As part of the exploratory data analysis (EDA), an initial cleaning process was performed to standardize empty or inconsistent values. Various representations of missing data—such as “NA”, “N/A”, “na”, “-”, empty strings, and whitespace—were replaced with pandas’ standard missing-value marker (pd.NA). This step enabled an accurate count of missing entries and simplified their treatment in later stages.

Using the functions `df.info()` and `df.shape`, a general characterization of the dataset was obtained, identifying the total number of records and features, number of non-null values per variable among other characteristics.

During this stage, inconsistencies were also found in the naming of particulate-matter variables, as some stations reported PM_{2.5} under labels such as `pm_2`, `pm2`, `PM2`, and others. To ensure consistency in the processing pipeline, a regular expression was applied to unify all variants under a standardized name (`pm2_5`). Subsequent verification confirmed that all labels had been normalized correctly.

Once inconsistencies were resolved, the time column was converted to the datetime format, and the PM_{2.5} series was ensured to be in a valid numeric type. Based on this temporal information, a monthly variable was generated and the monthly

average of PM_{2.5} was computed, allowing the identification of seasonal patterns and significant variations in the pollutant throughout the year. This analysis revealed a notable increase during the first months, likely explained by atmospheric conditions typical of that period (higher humidity, lower wind dispersion), combined with urban mobility dynamics.

To evaluate the data quality of each monitoring station, the number of valid observations per variable and station was calculated, including both pollutant data (PM₁₀, PM_{2.5}, NO₂, O₃) and meteorological variables (wind direction and speed, humidity, solar radiation, and precipitation). This comparative assessment identified the Santa Cruz – Girón Norte station as the one with the highest completeness and consistency in its records, making it the most suitable choice for predictive modeling. Since the project’s objective is not to compute an Air Quality Index but to predict PM_{2.5} concentration exclusively, the availability of the relevant variables in this station fully justified its selection.

Beyond the initial descriptive analysis, relationships between PM_{2.5} and key meteorological variables were explored. In particular, the following patterns were observed:

- **Inverse relationship with wind speed:** low wind speeds favor pollutant accumulation, while higher speeds promote dispersion.
- **Positive relationship with relative humidity:** elevated humidity levels appear to facilitate the persistence of fine particles in suspension.
- **Negative relationship with solar radiation:** higher radiation increases atmospheric turbulence and mixing, reducing PM_{2.5} concentrations.

Finally, boxplots by hour of the day revealed daily patterns consistent with humidity and radiation behavior: higher concentrations during early morning and nighttime hours, and lower concentrations during periods of high solar radiation. These results reflect the marked influence of atmospheric and temporal cycles on pollutant dynamics.

Overall, the EDA process enabled the cleaning of the dataset, the identification of the most suitable station for modeling, and a clear understanding of the relationships between PM_{2.5} and meteorological variables, thereby providing a solid foundation for the construction of the predictive model.

TRAINING

In the development of the air-quality prediction system, three supervised regression models were implemented. The selection of these algorithms responded to the need to evaluate different predictive approaches with distinct characteristics and capabilities—both individual and ensemble-based—in order to identify the model with the best generalization capacity for meteorological data.

The models used were:

- Random Forest Regressor
- AdaBoost Regressor (Adaptive Boosting)
- Ridge Regression (L2)

Random Forest Regressor

The Random Forest Regressor served as the primary model in this study. It is an ensemble method based on bagging, where multiple decision trees are trained independently using different subsets of data and features [7].

Key advantages:

- Reduction of overfitting present in a single tree
- Greater stability and robustness to noise
- Improved predictive performance
- Ability to estimate feature importance

This model successfully captured complex patterns in meteorological behavior, showing superior performance compared to an individual decision tree.

AdaBoost Regressor (Adaptive Boosting)

The AdaBoost Regressor is a boosting technique in which multiple weak learners (typically shallow trees) are trained sequentially. Each iteration adjusts the weight of errors made by previous models, progressively focusing on the most difficult cases [9].

Key advantages:

- Better handling of systematic errors
- Ability to enhance simple models
- Sensitivity to noise
- Higher risk of overfitting on unstable data

AdaBoost was implemented to evaluate an incremental learning approach and compare its performance against the bagging-based model.

Ridge Regression with L2 regularization

Ridge Regression offers several advantages when modeling time series such as hourly PM2.5 concentrations. By incorporating an L2 penalty, it reduces overfitting and improves generalization in the presence of noisy data or highly correlated variables, mitigating issues related to multicollinearity.

This regularization stabilizes the coefficients, prevents excessive magnitudes, and provides a more robust representation of the underlying linear relationship. Thanks to these properties, Ridge serves as a solid comparison baseline against more complex methods, offering consistent performance and a clear interpretative framework with a low risk of overfitting [8].

Key advantages:

- Allows explicit control of the regularization level via the hyperparameter λ , facilitating optimization of the bias–variance trade-off.
- Improves numerical stability during training, especially when the number of variables is large relative to the number of samples.
- Can effectively handle scenarios with few observations compared to the number of predictors.

EVALUATION

To ensure a fair and consistent comparison among the evaluated models, the three selected algorithms (Random Forest Regressor, AdaBoost Regressor, and Ridge Regression) were trained and tested using exactly the same dataset and the same train–test split. This guarantees that any difference in performance can be attributed solely to the inherent characteristics of each model, avoiding biases derived from variations in the data, sampling, or temporal distribution.

The evaluation of the predictive system was approached from two complementary perspectives. First, the performance metrics obtained by each model were analyzed, considering indicators such as RMSE, MAE, and R^2 , with the aim of determining the accuracy and stability of their predictions. Second, the computational complexity associated with each algorithm was assessed, particularly training and prediction times, since these factors are crucial for the practical implementation of the model in a desktop application operating in near real time.

Table 1 presents the comparative performance results of the three models. It includes RMSE, MAE, and R^2 values for both the training and test sets, allowing analysis not only of each algorithm’s overall accuracy but also of its generalization ability and behavior with respect to overfitting. This table constitutes the central component of the quantitative analysis of predictive performance.

Table 1. Comparative Performance Metrics

Model	Dataset	RMSE	MAE	R ²
Random Forest	Train	1.422	0.835	0.989
	Test	3.716	2.213	0.960
AdaBoost	Train	4.701	3.277	0.936
	Test	4.701	3.277	0.936
Ridge Regression	Train	3.621	3.277	0.936
	Test	4.701	3.277	0.936

Source: Authors

On the other hand, Table 2 summarizes the training and prediction times for each model, which are essential for assessing the computational feasibility of their implementation. This analysis is particularly relevant given the project's goal of integrating the final model into a lightweight desktop application, where temporal efficiency and fast response times are key components to ensure an adequate user experience.

Table 2. Computational Complexity

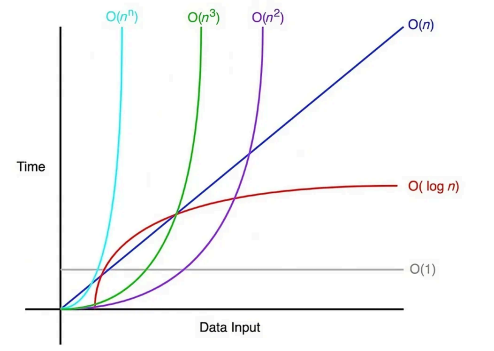
Model	Training Time (s)	Prediction Time (s)
Random Forest	33.5556	0.1548
AdaBoost	8.0908	0.0849
Ridge Regression	0.009441	0.002251

Source: Authors

Taken together, the results presented in both tables show that although Ridge Regression offers the lowest computational complexity, its accuracy is considerably lower than that of Random Forest. AdaBoost, while more efficient than Random Forest in execution time, exhibited intermediate performance and greater sensitivity to outliers. Random Forest, in contrast, achieved the best balance between accuracy, stability, and robustness, justifying its selection as the final model for the application. Its ability to generalize effectively and its tolerance to noise make it the most reliable option for hourly $PM_{2.5}$ prediction in operational scenarios.

In addition to the two approaches for model evaluation, we also performed an analysis of the computational complexity for each algorithm using the Big O Notation (O). The figure below illustrates various typical temporal complexity curves in machine learning algorithms, which helps to contextualize the training and prediction times observed in the evaluated models.

Image 1. Big O Notation (O)



Source: Medium (2022)

The complexity of Ridge Regression primarily depends on the amount of data (n) and the number of variables used (p). This model operates directly on matrices that include all features, so adding more variables slightly increases the training time. However, Ridge remains very fast, as reflected in the results: its training took only 0.009 s and prediction took 0.001 s, demonstrating that it maintains efficiency even with larger datasets. Its speed makes it ideal when an immediately executing linear model is required, although it doesn't capture complex relationships like AdaBoost or Random Forest.

For AdaBoost, the complexity is determined by the number of data points (n) and the number of weak trees (m). The number of variables (p) has little influence, as each tree is small and doesn't evaluate all features simultaneously. In our case, with $m = 200$, training took approximately 8 s and prediction took 0.084 s. This reflects that the computational load depends mainly on the number of trees, and in exchange, a greater capacity to capture non-linear patterns in the data is obtained.

Random Forest combines the amount of data (n), the number of trees (m), and the search depth in each tree ($\log n$), making it the most computationally expensive model. Each tree performs multiple splits to find the best cut points, and using $m = 200$, this process is repeated numerous times. Consequently, training took 33.55 s and prediction took 0.154 s. Despite its longer execution time, Random Forest offered the best performance in the evaluated metrics, thanks to its ability to handle complex and noisy relationships present in the $PM_{2.5}$ data.

Comparative Analysis

Finally, to select the most suitable model for implementation in the proposed desktop application, a qualitative comparative analysis was conducted based on performance, robustness, interpretability, and computational feasibility. These criteria allow for evaluating not only the numerical accuracy of each model but also its practical relevance in an operational environment where efficiency, stability, and the ability to handle complex relationships are essential. Table 3 summarizes this comparison, integrating the strengths and limitations of each approach.

Table 3. Comparison of Criteria for Model Selection

Criterion	Random Forest	AdaBoost	Ridge Regression
Predictive performance (RMSE/R²)	Best overall performance	Moderate	Limited compared to RF
Robustness to noise and outliers	High	Low-medium (sensitive to outliers)	Medium
Ability to model non-linear relationships	High	Medium	Low (linear model)
Train-test stability	High	High (but less accurate)	Medium
Interpretability	Medium (variable importance)	Low	High
Computational cost	Medium	Medium-low	Very low
Suitability for a lightweight desktop application	Yes: optimal balance between accuracy and speed	Possible but less accurate	Yes, but low accuracy
Final model suitability	✓ Selected	✗ No	✗ Baseline

Source: Authors

As shown in Table 3, the Random Forest Regressor offers the best balance between predictive performance, stability in the presence of noise, and the ability to capture non-linear relationships—key characteristics for an application intended for forecasting atmospheric pollutants. Although AdaBoost exhibited competitive execution times and Ridge Regression stood out for its low computational complexity, neither achieved the level of accuracy and robustness required to ensure reliable predictions in a real-world setting. For these reasons, and considering the need for a consistent and operational model for the desktop application, Random Forest was selected as the final model for the predictive system.

MODEL EXPORT AND IMPLEMENTATION

Once Random Forest was established as the optimal model, it was exported along with the exact set of columns used during training, thus ensuring consistency between the modeling and inference phases. This packaging generated the file `best_rf_pm25.pkl`, which was later integrated into the development of the AirAMB desktop application, designed as an operational tool for hourly PM_{2.5} prediction ($t+1$). To provide an intuitive and visually appealing user experience, a dedicated visual identity was created—including a logo and a coherent graphic style—and the interface elements were refined to prioritize clarity, accessibility, and usability.

To further evaluate the model in concrete situations, ten random timestamps from the test set were selected, and the Random Forest predictions were compared against the observed PM_{2.5}($t+1$) values. This punctual validation enabled the individual examination of absolute and relative errors, the identification of potential overestimation or underestimation patterns, and the verification of the model's coherence when confronted with real cases. Additionally, summary metrics such as MAE, RMSE, and average bias were computed to provide an additional measure of performance beyond the training environment.

For this purpose, the function `evaluar_instante` was defined, enabling the analysis of the model's behavior at a specific date and time within the station's historical record. Based on the provided timestamp, the function retrieves the corresponding entry, generates the PM_{2.5}($t+1$) prediction, and compares it with the observed value, reporting both absolute and relative error. It also prints all input variables used—PM_{2.5} and PM₁₀ with their lags, temporal features, and meteorological parameters—which facilitates reproducing the exact scenario inside the AirAMB desktop application and performing cross-validation between the notebook and the graphical interface. The results obtained through this process highlight the methodological rigor of the validation approach and its consistency with the graphical interface outputs.

Regarding the design and implementation of the application, in order to illustrate the visual identity developed for the application, both the logo and a portion of the graphical interface

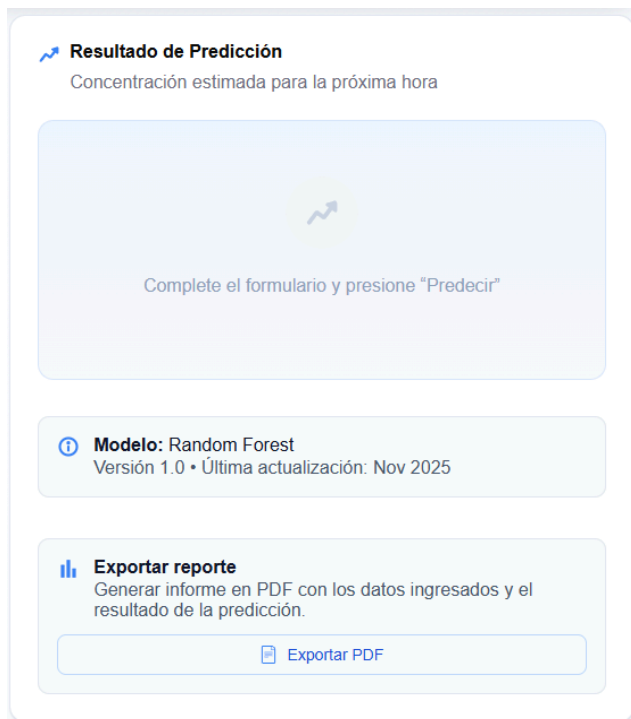
are presented. These elements showcase the selected color palette, the clean layout, and the emphasis on an intuitive and streamlined user experience. The visual design reinforces the operational nature of the tool while maintaining simplicity, readability, and a user-centered interaction flow.

Image 2. AirAMB Logo



Source: Authors

Image 3. Results Interface and Report Generation



Source: Authors

The application was built using PyWebview, enabling the combination of a modern interface based on HTML and CSS with native Python execution of the model. From a software engineering perspective, the application was structured into dedicated scripts, each responsible for a specific component of the workflow—configuration handling, feature processing, $PM_{2.5}$ prediction, validation routines, and the main execution logic. This

modular design not only improves maintainability but also helps isolate essential operations. For example, although the model uses transformed temporal features such as $hour_sin$ and $hour_cos$, it is not meaningful to request these values from the user. Instead, a specific script automatically computes and injects these variables internally before sending the input to the model, preserving both usability and predictive coherence. The overall structure reflects a deliberate effort to integrate a machine-learning model into a functional, user-oriented desktop environment while keeping the interface intuitive, clean, and aligned with the visual identity defined during the design phase.

After development, the project was packaged using PyInstaller to generate a standalone executable, and finally distributed through an installer created with Inno Setup, ensuring a simple and professional installation process for end users. The initial interface prototype was designed in Figma.

Main functionalities of AirAMB include:

- Allowing users to input pollutant concentrations, temporal variables, and meteorological data to generate predictions.
- Displaying the estimated $PM_{2.5}(t+1)$ concentration immediately.
- Indicating the model used and the implemented version.
- Offering the option to export a PDF report containing the input data and the prediction result.
- Providing an intuitive, organized, and easy-to-interpret interface.

CONCLUSIONS

The exploratory analysis made it possible to identify clear patterns in the behavior of $PM_{2.5}$ both throughout the year and within the daily cycle. A significant increase in the pollutant was observed during the first months of the year and in the early morning hours—an expected phenomenon given the typical atmospheric conditions of that period: lower solar radiation, high humidity, and limited vertical mixing, all of which favor the accumulation of fine particulate matter.

The Santa Cruz – Girón Norte station was selected as the basis for modeling because it showed the highest completeness and consistency in its records. This data availability allowed the models to be trained with greater stability, reducing uncertainty associated with gaps or inconsistencies and producing more robust results compared to the other stations evaluated.

The analysis confirmed that the recent dynamics of $PM_{2.5}$ play a decisive role in its future evolution. The incorporated lags showed a strong relationship with the value to be predicted, highlighting the autocorrelated nature of the pollutant. Likewise, the cyclical encoding of temporal variables enabled a more accurate representation of daily and weekly cycles, avoiding artificial breaks between consecutive hours or days.

Among the evaluated techniques, tree-based models—especially Random Forest—achieved the best performance, consistent with their ability to handle non-linear relationships, interactions between variables, and noisy data. In contrast, linear models such as Ridge Regression showed structural limitations when facing more complex patterns found in environmental data.

The temporal validation strategy, in which training is performed on an initial period and testing on subsequent observations, was essential for obtaining realistic results. This methodology prevented information leakage from the future into the training process and allowed for evaluating the model's ability to predict unseen conditions, faithfully replicating the system's operational use.

Although the results were satisfactory, there remains room for improving prediction accuracy. Incorporating advanced models such as LSTMs, performing more rigorous outlier cleaning, or integrating higher-resolution satellite and meteorological information could strengthen predictive capacity, especially in scenarios where air quality changes more rapidly.

Overall, the project demonstrates that machine-learning techniques are a valuable tool for operational $PM_{2.5}$ prediction in Bucaramanga. The adopted approach allows anticipating increases in the pollutant and provides meaningful support for air-quality management and the formulation of preventive strategies.

REFERENCES

- [1] D. A. Franco Pineda, Analysis and Characterization of PM_{10} and $PM_{2.5}$ Particulate Matter in the City of Manizales. Universidad Nacional de Colombia, 2020. Available: <https://bffrepositorio.unal.edu.co/server/api/core/bitstreams/4d9ceee1-5e9d-4adb-b476-d5b6192f9f76/content>
- [2] R. Yu, Y. Yang, L. Yang, G. Han, and O. A. Move, "RAQ: A Random Forest Approach for Predicting Air Quality in Urban Sensing Systems," *Sensors*, vol. 16, no. 1, p. 86, 2016.
- [3] X. Yi, J. Zhang, Z. Wang, T. Li, and Y. Zheng, "Deep Distributed Fusion Network for Air Quality Prediction," in *Proc. ACM SIGKDD Conf. Knowledge Discovery and Data Mining*, 2018.
- [4] K. Veljanovska and A. Dimoski, "Air Quality Index Prediction Using Simple Machine Learning Algorithms," *Int. J. Emerging Trends Technol. Comput. Sci.*, vol. 7, pp. 25–30, 2018.
- [5] United Nations Environment Programme (UNEP), "Air Pollution: General Information," 2024. Accessed: Nov. 28, 2025. Available: <https://www.unep.org/es/explore-topics/air/about-aire>
- [6] G. Raimondo, A. Montuori, W. Moniaci, E. Pasero, and E. Almkvist, "A Machine Learning Tool to Forecast PM_{10} Level," in *Proc. 5th Conf. Artificial Intelligence Applications to Environmental Science (AMS)*, San Antonio, TX, USA, 2007.
- [7] Scikit-Learn Documentation, "RandomForestRegressor," 2024. Available: <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html>
- [8] IBM, "Ridge Regression: What It Is, How It Works, and What It Is Used For," IBM Think, 2024. Available: <https://www.ibm.com/es-es/think/topics/ridge-regression>
- [9] S. Baladram, "AdaBoost Classifier, Explained: A Visual Guide with Code Examples – Putting the weight where weak learners need it most," Medium, Nov. 10, 2024. Available: <https://medium.com/data-science/adaboost-classifier-explained-a-visual-guide-with-code-examples-fc0f25326d7b>
- [10] Área Metropolitana de Bucaramanga (AMB), Reporte Calidad de Aire AMB [Dataset], 2024. Available: https://www.datos.gov.co/Ambiente-y-Desarrollo-Sostenible/Reporte-Calidad-de-Aire-AMB/58ct-h586/about_data